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Web Technology

Quality Laboratory Manual

ExperimentNo. 08

Design web form and validate it using regular expressions



Course Instructor–
Mrs.Amruta R.Patil
ASSISTANT PROFESSOR

Experiment No.08

Title of Experiment: Design web form and validate it using regular expressions

Aim of Experiment: To design a web form using HTML and implement input validation using Regular Expressions to ensure correct user input.

System Requirements –

Operating System: Windows 10 or above

RAM: Minimum 4 GB

Software/s Needed for Experiment-

- Web Browser
- Text Editor / VS Code
- XAMPP / WAMP Server (Optional for PHP validation)

Experiment Objectives:

- To understand the concept of web forms in HTML.
- To learn form input validation techniques.
- To implement Regular Expressions (Regex) for validating form fields.
- To validate fields such as name, email, phone number, and password.
- To ensure correct user input before form submission.

Experiment Outcomes:

- Design an HTML registration form.
- Apply regular expressions for input validation.
- Prevent incorrect or invalid data entry.
- Improve data accuracy and form security.

Theory:

A Web Form is used to collect user input on a web page. HTML provides different form elements such as:

- <input>
- <textarea>
- <button>

Form validation ensures that users enter correct and properly formatted data before submission.

Regular Expressions (Regex)

A Regular Expression is a sequence of characters used to define a search pattern. It is commonly used for input validation.

Examples:

Field	Regular Expression	Description
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Field	Regular Expression	Description
Name	[A-Za-z]{3,}	Only letters, minimum 3 characters
Email	[a-z0-9._%+~]+@[a-z0-9.-]+\.[a-z]{2,}	Valid email format
Phone	[0-9]{10}	Exactly 10 digits
Password	(?=.*[A-Z])(?=.*[0-9]).{6,}	Minimum 6 characters with number and capital letter

Regular expressions help maintain data integrity and validation in web forms.

Program

HTML Form with Regular Expression Validation

```
<!DOCTYPE html>
<html>
<head>
<title>Student Registration Form</title>
</head>

<body>

<h2>Student Registration Form</h2>

<form>

Name:<br>
<input type="text" name="name" pattern="[A-Za-z]{3,}"
title="Name must contain only letters and minimum 3 characters" required>
<br><br>

Email:<br>
<input type="email" name="email"
pattern="[a-z0-9._%+~]+@[a-z0-9.-]+\.[a-z]{2,}$"
title="Enter a valid email address" required>
<br><br>

Phone Number:<br>
<input type="text" name="phone"
pattern="[0-9]{10}"
title="Enter 10 digit phone number" required>
<br><br>

Password:<br>
<input type="password" name="password"
pattern="(?=.*[A-Z])(?=.*[0-9]).{6,}"
title="Password must contain at least 6 characters, one number and one uppercase letter" required>
<br><br>

<input type="submit" value="Submit">

</form>

</body>
</html>
```

Task for Students: Create a Student Registration Web Form and implement validation using Regular Expressions.

Task Requirements:

- Design an HTML web form with at least 5 input fields.
- Include fields such as Name, Email, Phone Number, Password, and Address.
- Apply regular expression validation using the pattern attribute.
- Display validation messages for incorrect input.
- Ensure the form does not submit invalid data.

Submission Requirement:

- HTML Form Code
- Screenshot of Web Form
- Screenshot showing validation error message
- Screenshot of successfully submitted form

Observations:

In this experiment, an HTML web form was designed to collect student registration details. Regular expressions were applied using the pattern attribute to validate fields like name, email, phone number, and password. The browser automatically displayed validation messages when incorrect input was entered. This ensured that only valid and properly formatted data was submitted through the form.

Conclusion:

This experiment demonstrated how web forms can be designed and validated using Regular Expressions. They learned how to prevent invalid input and ensure accurate data collection, which is essential for developing secure and reliable web applications.

Expected Oral Questions:

1. What is a web form?
2. What is form validation?
3. What is a regular expression?
4. What is the use of the pattern attribute?
5. What is the difference between client-side and server-side validation?
6. What is the use of the required attribute?
7. How are regular expressions used in web development?

FAQs in Interview:

1. What are regular expressions used for?
2. What is the difference between HTML validation and JavaScript validation?
3. What is the syntax of a regular expression?
4. What are the advantages of form validation?

5. Can regular expressions validate complex data formats?
6. What happens if form validation is not implemented?
7. What are common validation techniques used in web forms?
8. How does validation improve web application security?